

Yao-Lee-Kondo 模型的 Majorana 重新推导

1. 传导电子算符定义

将传导电子分成四个 majorana 算符

$$\begin{aligned}c_{j\uparrow} &= \frac{1}{2}(\eta_{1\uparrow}^j - i\eta_{2\uparrow}^j), & c_{j\downarrow} &= \frac{1}{2}(\eta_{1\downarrow}^j - i\eta_{2\downarrow}^j) \\c_{j\uparrow}^\dagger &= \frac{1}{2}(\eta_{1\uparrow}^j + i\eta_{2\uparrow}^j), & c_{j\downarrow}^\dagger &= \frac{1}{2}(\eta_{1\downarrow}^j + i\eta_{2\downarrow}^j)\end{aligned}\tag{1}$$

在此约定下，占据数算符为：

$$n_{j\uparrow} = c_{j\uparrow}^\dagger c_{j\uparrow} = \frac{1}{4}(\eta_{1\uparrow}^j + i\eta_{2\uparrow}^j)(\eta_{1\uparrow}^j - i\eta_{2\uparrow}^j) = \frac{1}{2} - \frac{i}{2}\eta_{1\uparrow}^j\eta_{2\uparrow}^j\tag{2}$$

2. 传导电子自旋 $\vec{s}_j$ 的推导

利用自旋算符标准定义 $s^a = \frac{1}{2}c^\dagger\sigma^a c$ ：

• **Z 分量:**

$$\begin{aligned}s_j^z &= \frac{1}{2}(n_{j\uparrow} - n_{j\downarrow}) \\&= \frac{1}{2}\left[\left(\frac{1}{2} - \frac{i}{2}\eta_{1\uparrow}^j\eta_{2\uparrow}^j\right) - \left(\frac{1}{2} - \frac{i}{2}\eta_{1\downarrow}^j\eta_{2\downarrow}^j\right)\right] \\&= -\frac{i}{4}(\eta_{1\uparrow}^j\eta_{2\uparrow}^j - \eta_{1\downarrow}^j\eta_{2\downarrow}^j)\end{aligned}\tag{3}$$

• **X 分量:**

$$\begin{aligned}s_j^x &= \frac{1}{2}(S_j^+ + S_j^-) \\&= \frac{1}{2}(c_{j\uparrow}^\dagger c_{j\downarrow} + c_{j\downarrow}^\dagger c_{j\uparrow}) \\&= \frac{1}{2}\left[\frac{1}{4}(\eta_{1\uparrow}^j + i\eta_{2\uparrow}^j)(\eta_{1\downarrow}^j - i\eta_{2\downarrow}^j) + \frac{1}{4}(\eta_{1\downarrow}^j + i\eta_{2\downarrow}^j)(\eta_{1\uparrow}^j - i\eta_{2\uparrow}^j)\right] \\&= \frac{1}{8}[\{\eta_{1\uparrow}^j, \eta_{1\downarrow}^j\} + \{\eta_{2\uparrow}^j, \eta_{2\downarrow}^j\} + i[\eta_{2\uparrow}^j, \eta_{1\downarrow}^j] - i[\eta_{1\uparrow}^j, \eta_{2\downarrow}^j]] \\&= -\frac{i}{4}(\eta_{1\uparrow}^j\eta_{2\downarrow}^j - \eta_{2\uparrow}^j\eta_{1\downarrow}^j)\end{aligned}\tag{4}$$

• **Y 分量:**

$$\begin{aligned}
s_j^y &= \frac{1}{2i}(S_j^+ - S_j^-) \\
&= \frac{1}{2i}(c_{j\uparrow}^\dagger c_{j\downarrow} - c_{j\downarrow}^\dagger c_{j\uparrow}) \\
&= \frac{1}{8i}[(\eta_{1\uparrow}^j + i\eta_{2\uparrow}^j)(\eta_{1\downarrow}^j - i\eta_{2\downarrow}^j) - (\eta_{1\downarrow}^j + i\eta_{2\downarrow}^j)(\eta_{1\uparrow}^j - i\eta_{2\uparrow}^j)] \\
&= \frac{1}{8i}[[\eta_{1\uparrow}^j, \eta_{1\downarrow}^j] - i\{\eta_{1\uparrow}^j, \eta_{2\downarrow}^j\} + i\{\eta_{2\uparrow}^j, \eta_{1\downarrow}^j\} + [\eta_{2\uparrow}^j, \eta_{2\downarrow}^j]] \\
&= -\frac{i}{4}(\eta_{1\uparrow}^j \eta_{1\downarrow}^j + \eta_{2\uparrow}^j \eta_{2\downarrow}^j)
\end{aligned} \tag{5}$$

3. Yao-Lee 局域自旋 $\vec{S}_j$ 的表示

为了保持 $S = \frac{1}{2}$ 代数且满足 $\chi^2 = 1$ ，局域自旋定义为：

$$\vec{S} = -\frac{i}{2}\vec{\chi} \times \vec{\chi} \tag{6}$$

$$S_j^x = -\frac{i}{2}\chi_j^y \chi_j^z, \quad S_j^y = -\frac{i}{2}\chi_j^z \chi_j^x, \quad S_j^z = -\frac{i}{2}\chi_j^x \chi_j^y \tag{7}$$

4. 最终近藤耦合项 H_K

将 $\vec{s}_j$ 与 $\vec{S}_j$ 代入 $H_K = J \sum_j \vec{S}_j \cdot \vec{s}_j$ 。由于 S^a 和 s^a 均带有 $-i$ 系数，其乘积项出现 $(-i)^2 = -1$ ：

$$\begin{aligned}
H_K &= -\frac{J}{8} \sum_j [\chi_j^x \chi_j^y (\eta_{1\uparrow}^j \eta_{2\uparrow}^j - \eta_{1\downarrow}^j \eta_{2\downarrow}^j) \\
&\quad + \chi_j^y \chi_j^z (\eta_{1\uparrow}^j \eta_{2\downarrow}^j - \eta_{2\uparrow}^j \eta_{1\downarrow}^j) \\
&\quad + \chi_j^z \chi_j^x (\eta_{1\uparrow}^j \eta_{1\downarrow}^j + \eta_{2\uparrow}^j \eta_{2\downarrow}^j)]
\end{aligned} \tag{8}$$